UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PROGRAMMING I
COURSE CODE	CSC402
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

PART A (10 MARKS)

Answer ALL questions.

1. Which declaration creates a floating-point variable named price?
   A. float price;
   B. int price;
   C. bool price;
   D. char price;

2. Which structure chooses between two alternative paths?
   A. array
   B. struct
   C. for
   D. if...else

3. Which identifier is valid in C++?
   A. total score
   B. int
   C. total_score
   D. 2total

4. Which values can a bool variable hold?
   A. any character
   B. true or false
   C. any decimal
   D. any integer

5. Which loop is commonly used to process every array element?
   A. break
   B. if
   C. for
   D. switch

6. What does a compiler do?
   A. Draws flowcharts automatically
   B. Executes one source statement at a time
   C. Translates the whole source program before execution
   D. Stores only integer values

7. Which tool represents an algorithm using standard graphical symbols?
   A. Compiler
   B. Pseudocode
   C. Debugger
   D. Flowchart

8. Which statement is most suitable for selecting among several constant cases?
   A. do...while
   B. while
   C. switch
   D. for

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9. How many times does a while loop execute when its condition is false initially?
   A. Until the user stops it
   B. Two
   C. Zero
   D. One

10. Which header is required for cin and cout?
   A. <cmath>
   B. <iomanip>
   C. <iostream>
   D. <string>

PART B (60 MARKS)

QUESTION 1

This question tests algorithms and programming errors.

a) Define algorithm and pseudocode.

(2 marks)

b) Explain THREE types of programming errors and give one example of each.

(3 marks)

QUESTION 2

A program reads a positive integer N and calculates the sum and average of the numbers 1 to N.

a) Draw a flowchart for the sum and average calculation.

(6 marks)

b) Write pseudocode and a complete C++ program for the algorithm.

(7 marks)

QUESTION 3

This question tests operator precedence and assignment.

a) Evaluate the following expressions: (i) 7 + 3 * 2, (ii) (7 + 3) * 2, (iii) 17 % 5, (iv) 17 / 5, (v) 17.0 / 5.

(5 marks)

b) Write C++ statements to declare two integers a and b, read their values, and display their sum, difference, product and remainder.

(6 marks)

QUESTION 4

Study the following nested loop:

```
for (int i = 1; i <= 3; i++) {
   for (int j = 1; j <= i; j++) {
      cout << j;
   }
   cout << endl;
}
```

a) Trace the program and show the exact output.

(10 marks)

b) Identify TWO logic errors in a version that prints four rows but reuses j without resetting it. Rewrite the corrected code.

(5 marks)

QUESTION 5

A shop stores the prices of five items in an array.

a) Declare an array named prices with the values 20.5, 55.0, 15.9, 75.0 and 10.0. Display all values.

(10 marks)

b) Write a program segment that applies a 10% discount to prices above RM50 and stores the results in a second array. Display both arrays.

(6 marks)

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PART C (30 MARKS)

QUESTION 1

A bookstore records up to five books. Each book has a title, price and quantity in stock.

Define a struct Book. Write a complete C++ program that reads the records, calculates the total inventory value, and displays the title of the most expensive book.

(15 marks)

QUESTION 2

A bookstore records up to five books. Each book has a title, price and quantity in stock.

Define a struct Book. Write a complete C++ program that reads the records, calculates the total inventory value, and displays the title of the most expensive book.

(15 marks)

END OF QUESTION PAPER